Calibrated Analog Joystick

Preserve raw evidence, shape two axes, debounce one selection

Three observations; one deterministic snapshot

Lesson	031 — component	**Time**	150–210 minutes
Prerequisites	Lessons 002, 007, and 008	**Board**	Arduino Mega 2560
Evidence	TP-X, TP-Y, RGB preview	**Status**	host verified; bench open
Energy class	E1: USB, passive joystick, resistor-limited RGB LED		

Specimen gate. The official Elegoo manifests authorize a "Joystick module" family, not a universal pin order or catalog identity. Before wiring, read the labels on the exact unpowered board and establish its supply rating and signal roles. Do not infer KY-023, pin order, polarity, or axis direction from board color or an internet photograph. Unknown markings stop the physical procedure; they do not block the host lesson.

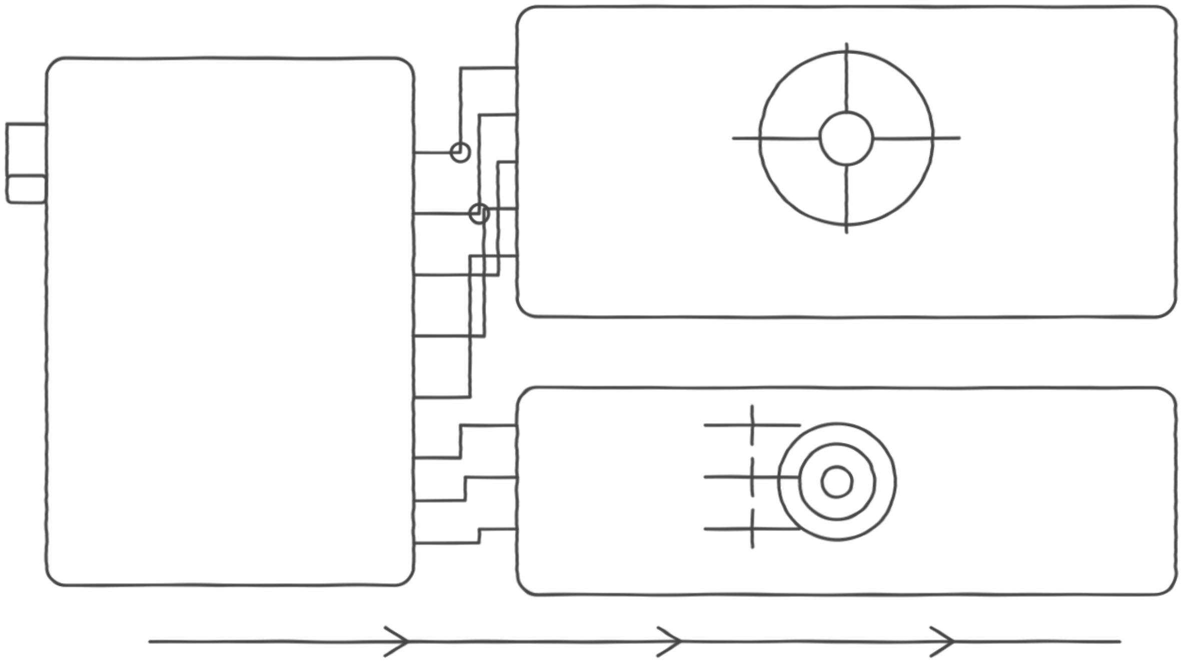
1 Mission and learning outcomes

X raw + Y raw + switch raw → calibrated snapshot → RGB direction and magnitude.

By the end, you can:

- distinguish raw ADC evidence from normalized position;
- map asymmetric travel around a measured center without floating point;
- explain why a center dead zone is not discarded evidence;
- debounce selection with caller-supplied time;
- predict initialization rollback and reverse-order shutdown;
- prove acquisition separately from input-mode safe state.

2 Orientation and unpowered inspection



Alternative text: monochrome orientation plate showing an inspected joystick module connected to Mega A0, A1, D22, 5 V, and ground; TP-X and TP-Y at the two analog signals; and a separately resistor-limited common-cathode RGB LED on D5, D6, and D7.

Observed role	Reference pin	Connection after inspection
X signal: VRx, X, or HOR	A0 / D54	module signal; TP-X to Mega GND
Y signal: VRy, Y, or VER	A1 / D55	module signal; TP-Y to Mega GND
select: SW, KEY, or SEL	D22	switch-to-ground, internal pull-up
supply and common	5 V and GND	only after rating and continuity checks
RGB preview	D5, D6, D7	each pin through 330 Ω to LED channel

Unpowered checklist.

☐ record every module label and its physical order;
☐ continuity-check the ground terminal and select contact;
☐ verify three separate RGB resistors and LED common polarity;
☐ verify TP-X and TP-Y cannot contact 5 V or adjacent rows;
☐ confirm no motor, relay, external supply, or unknown module.

Exact module/revision: ____________________ Supply source/rating: ____________________.

3 Calibration contract

Each axis stores five facts: analog pin, observed minimum, center, observed maximum, and dead-zone half-width. Valid calibration requires

$$m < c < M, \qquad c - d > m, \qquad c + d < M.$$

Here m is the observed minimum, c the center, M the observed maximum, and d the dead zone. These are configured observations, not claims about universal joystick travel.

$$p(r) = \begin{cases} -1000\,\dfrac{c-d-r}{c-d-m}, & r < c-d, \\ 0, & c-d \le r \le c+d, \\ 1000\,\dfrac{r-(c+d)}{M-(c+d)}, & r > c+d. \end{cases}$$

The implementation clamps before narrowing, uses 32-bit intermediates, and applies inversion last. A raw value at or beyond an observed endpoint sets `saturated`; it does not assert an electrical fault. Raw values remain in the snapshot even when their normalized position is zero.

3.1 Prediction worksheet

Use $m = 80$, $c = 520$, $M = 960$, and $d = 40$.

Raw count	Region prediction	Position prediction	Saturated?
80	__________	__________	________
479	__________	__________	________
480	__________	__________	________
560	__________	__________	________
561	__________	__________	________
960	__________	__________	________

4 Lifecycle, ownership, and sample order

Construction performs no hardware operation. Initialization validates both axis configurations, three distinct pins, and Mega capabilities before claiming X, then Y, then select. Any failure releases the claims from that attempt in reverse order. Repeated successful initialization is inert.

Every `update(now)` reads X once, Y once, then delegates exactly one select sample to the existing `Button`. Its snapshot contains raw and calibrated axes, raw and debounced selection levels, press and release events, and one complete status. Events remain stable until the next update.

```
const adk::Status status = joystick.update (now);
const adk::AnalogJoystickSnapshot view =
    joystick.snapshot ();

showJoystickPosition (view);
showSelectionEvent   (view);
```

Shutdown stops select, Y, then X. It preserves the last raw and interpreted evidence. The retained snapshot reports a not-initialized status. The three input endpoints return to `INPUT`; an RGB presenter is a separate owner and must also be shut down by the sketch.

5 Circuit-native presentation

The RGB LED is not part of `AnalogJoystick`. The narrative sketch maps the stable snapshot to an independent presentation:

- a short blue pulse after all resources initialize proves acquisition;
- red intensity represents negative X, green intensity positive X;
- centered X uses blue at the same Y-derived brightness;
- Y scales the chosen direction's brightness;
- a debounced select event produces a brief all-channel marker.

The same color never carries the only distinction: TP-X and TP-Y retain direct voltage evidence, and the lesson names the selected channel and magnitude in text. Serial raw counts are optional supporting evidence.

5.1 Current worksheet

One 330 Ω resistor is required per RGB channel. The sketch commands at most one steady direction channel; the short event marker may command three. A conservative zero-forward-voltage bound is

$$I_{\text{channel}} \leq \frac{5.0\ \text{V}}{330\ \Omega} = 15.2\ \text{mA}, \qquad I_{\text{RGB,total}} \leq 45.5\ \text{mA}.$$

This is a design bound, not a measurement.

Quantity	Predicted	Measured	Interpretation
TP-X at rest	within 0–5 V	________	________________
TP-Y at rest	within 0–5 V	________	________________
each RGB resistor	330 Ω	________	________________
select released/pressed	high / low	________	________________

6 Narrative Mega example

```
void loop ()
{
    const adk::TimePoint now (millis ());

    observeJoystick (now);
    decidePreview   ();
    showPreview     (now);
}
```

Objects appear in dependency order: Mega runtime, resource registry, joystick, RGB endpoints, then presenter state. Setup reads as acquire, configure, start. The loop reads as observe, decide, actuate. It takes one joystick snapshot and never reads a pin directly.

7 Predict, observe, interpret

1. With power absent, predict the center-region TP-X and TP-Y voltages and the select levels. Record the exact inspected pin labels.
2. Apply USB power. Observe the blue acquisition pulse before interpreting any direction preview.
3. Hold the stick at rest. Record raw counts and TP-X/TP-Y. Choose center and dead-zone values from repeated observations, not a single sample.
4. Move X toward each mechanical limit without force. Compare raw count, normalized sign, direction channel, and TP-X.
5. Repeat for Y and explain how Y changes magnitude without changing X direction.
6. Bounce and hold select. Compare raw level, debounced level, and the single press/release events.

7. Invoke shutdown. Observe RGB inactivity separately from all three input pins returning to their documented modes; then remove USB.

8 Diagnosis and stop conditions

Observation	Stop, inspect, and interpret
label or pin order differs	remove USB; follow the exact board, not this drawing
TP-X or TP-Y outside rails	remove USB; inspect supply, ground, and signal role
axes exchanged	correct role-to-pin configuration; do not rename raw evidence
rest drifts outside dead zone	repeat observations; do not hide faults with a huge zone
direction sign is opposite	use explicit inversion after confirming wiring
multiple selection events	inspect debounce trace and supplied timestamps
preview but no blue pulse	do not claim acquisition; inspect startup sequence
hot part or unstable USB	remove USB immediately; troubleshoot unpowered

9 Exercises

1. Derive the two denominators for an asymmetric axis and explain why one global slope would bias the center.
2. Identify every invalid configuration that can be rejected before a hardware read or claim.
3. Draw the reverse rollback order when Y acquisition fails, then when select acquisition fails.
4. Explain why `saturated` preserves evidence but cannot diagnose a short circuit.
5. Treat the authorized tilt-ball family as an ordinary `DigitalInput` exercise. State why it does not justify a new semantic component.

10 Open hardware acceptance record

Software compilation and deterministic tests are not physical results. Leave every line blank until the named specimen and instrument are actually used.

Board/revision	____________	Commit	____________
Joystick markings	____________	RGB/resistors	____________
Supply/instrument	____________	Tool versions	____________
Operator/date	____________	Reviewer	____________

☐ exact labels, pin order, voltage rating, continuity, and unpowered wiring recorded;
☐ acquisition pulse predicted, observed, and interpreted;
☐ TP-X and TP-Y center, both limits, and calibration recorded;
☐ raw, normalized, centered, saturated, and inverted cases reconciled;
☐ select bounce, press, release, and stuck state observed;
☐ shutdown input modes and RGB inactivity established as separate evidence;
☐ USB power-removal result recorded.

Prediction: ____

Observation: ____

Interpretation and limits: ____

Stop rule. Remove USB power for unknown markings, uncertain pin order, an out-of-rail test point, missing resistor, wrong LED polarity, unexpected behavior, a hot part, unstable power, or disagreement between the schematic and breadboard. Do not move wiring while powered.

11 Reproduce the software evidence

```
make analog-joystick-test
make host-test-exceptions
make arduino-Lesson031AnalogJoystick
make size-check
make lessons
make lessons-check
make site-check
make hardware-card LESSON=031 BOARD=mega2560
```

Record the exact AVR core, compiler, measured flash, and static RAM with the canonical sketch. Re-measure when source or toolchain identity changes.

12 Reference trail

- ADK `analog_joystick.h`, `analog_input.h`, `button.h`, focused host tests, and canonical Lesson 031 sketch;
- ADK development, testing, safety, packaging, and PDF policies;
- the authorized Elegoo-set manifest for family provenance only;
- the exact module's markings and primary electrical source before powered work;
- Arduino Mega 2560 Rev3 datasheet for the board profile.

The searchable HTML reference is the primary accessible route. This printable PDF uses real text, headings, lists, and monochrome graphics but does not claim PDF/UA conformance.